MD SIMUL ISLAM

Software Engineering Student | AI, Multi-Agent Systems & Intelligent Software

Chengdu, Sichuan, China | +86 15680416508 | 23320120006@stu.nsu.edu.cn | WeChat: simul49

www.simul49.com | https://www.linkedin.com/in/simul49

SUMMARY

I'm a fourth-year Software Engineering student who enjoys turning ideas into practical technology. With three years of hands-on experience across software development, AI, and intelligent systems, I'm eager to learn, adapt, take responsibility, work smart, and contribute meaningfully to research and collaborative projects. And research interests include llms, Multi-Agent Systems, Self-Healing Distributed Systems, AIOps, AI Software Engineering, Digital Twins, and AI for science and future technologies.

EDUCATION

Chengdu Neusoft University **Chengdu, Sichuan, China**

Bachelor of Science in Software Engineering *Sep 2023 - Jun 2027 (Expected)*

• Overall Academic Average: 86.7%

• Relevant Coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks, Software Engineering, Software Architecture, Software Testing, Object-Oriented Programming, Web and Mobile Application Development, Microservices Development, Artificial Intelligence, Statistics & Probability, and Human–Computer Interaction

• Scholarship: Belt and Road Scholarship - Fully Funded Undergraduate Scholarship (4 Years).

RESEARCH EXPERIRNCR

1. Title: "LLM Multi-Agent Systems for Self-Healing Distributed Systems: A Critical Review of Architectures, Operational Autonomy and Evaluation Evidence" Research Status: Completed Manuscript / Under Review

• Conducting a critical review of 70+ LLM-based autonomous systems across the full self-healing lifecycle: detection, localisation, diagnosis, remediation proposal, validation, and execution

• Auditing multi-agent topologies (hierarchical supervisors, planner–executor designs, and single-agent configurations) to assess operational autonomy boundaries and safeguard mechanisms.

• Investigating systemic execution failure modes, including unconstrained action loops, unverified rollbacks, and lack of human escalation in critical operations

• Identifying empirical evaluation gaps in benchmark practices (e.g., SREGym, ITBench, OpenRCA) and proposing a normative framework with frozen backbones, token-cost metrics, and multi-agent ablation baselines.

Focus: LLMs | Multi-Agent Systems | AIOps | Self-Healing Systems | Distributed Systems Reliability |more
Abstract: https://hcn6bcdoy0v3.feishu.cn/docx/XZ0wdVcmnorA71xYWneckfFpnqd?from=from_copylink

2. *Title: "Privacy-Preserving Ambient Digital Twins for In-Home Elderly Fall Prevention: Real-Time Impact Detection and Prediction of Pre-Fall Frailty"* - Status: Ongoing - Conference Paper in Preparation

• Developing a privacy-preserving Ambient Digital Twin framework for real-time elderly fall detection and pre-fall frailty monitoring.

• Integrating AI, multimodal sensing, edge computing, and intelligent agents for continuous in-home health monitoring.

• Investigating real-time impact detection and pre-fall risk prediction to support proactive intervention and personalized care.

• Addressing privacy, reliability, and real-time processing challenges in AI-driven healthcare environments.

Focus: Digital Twins | AI | Edge AI | Multimodal Sensing | Elderly Care | Fall Detection | Healthcare AI

RESEARCH & TECHNICAL PROJECTS

MedTwin AI - AI-Driven Healthcare Digital Twin **Sep 2026 - Present**

Team Lead | AI Agents | Digital Twin | Healthcare AI | Software Development

Contributing to system architecture, software development, AI integration, and technical implementation in a 10-week project challenge with my classmates, while serving as the leader of my team.

AetherSwarm - Edge-Native Decentralized Swarm Robotics Platform **July 2026 - Aug 2026**

Python | Multi-Agent Systems | Swarm Robotics | Edge AI | P2P Networking | FastAPI | WebSocke

Implemented peer discovery, distributed task allocation, shared spatial intelligence, edge-based decision-making, fault-tolerant collaboration, multi-agent simulation, and real-time swarm monitoring.
https://github.com/simul49/AetherSwarm

Meta-Agent : Continuous Self-Evolving Prompt & Tool Optimizer **May 2026 - July 2026**

Python | LangGraph | Multi-Agent Systems | FastAPI | Vue.js | LLMs

Independently designed and developed an autonomous multi-agent pipeline for failure analysis, prompt and tool optimization, safety auditing, and validation. Integrated LangGraph, Agno, CrewAI, and AutoGen with multiple LLM engines, implementing sandbox testing, automated deployment, version tracking, rollback, and real-time monitoring through a Vue.js dashboard.

https://github.com/simul49/Autonomous-Decentralized-Agent-Consensus-State-Ledger-ADAC-SL-

TECHNICAL SKILLS

- Programming: Python, Java, JavaScript
- AI/ML: PyTorch, TensorFlow, Hugging Face, LangChain, LLMs, RAG, YOLO, OpenCV
- Agentic AI: LangGraph, CrewAI, AutoGen, Agno, Multi-Agent Systems
- Web & Software Development: Spring Boot,Vue.js, Django, FastAPI, REST APIs, WebSocket, MQTT
- Data & Automation: SPSS, Minitab, Power BI, n8n
- Tools & Infrastructure: GitHub, Linux, Docker, MySQL

PROFESSIONAL EXPERIENCE

Renata Limited **Gazipur, Dhaka, Bangladesh**

IT Specialist & Team Leader *Oct 2022 - Jul 2023*

- Collaborated with international engineering teams from China, Italy, and Egypt on production management, technical coordination, and operational support.
- Supported technical troubleshooting and coordinated team activities in a cross-functional working environment.
- Developed leadership, technical problem-solving, communication, and cross-cultural collaboration skills through international teamwork

SELECTED ACADEMIC PROGRAMS & INTERNATIONAL ACTIVITIES

- Tsinghua Shenzhen International Graduate School - Invited Participant, Tsinghua Peer Experience Day (2026)
- Beijing Institute of Technology - Selected Participant, Winter AI Program, 2025
- Tsinghua University - Selected Participant, HTGR Summer School (2025).
- Renmin University of China -Selected Participant, Summer Program (2025).
- Xi'an Jiaotong University -Selected Participant, Summer Program (2025).

AWARDS & ACTIVITIES

- Third Prize (Regional Level), 2026 iCAN College Student Innovation Competition: "Yixue Cup" Cross-Border Digital Marketing Challenge (Sichuan Division)
- HarmonyOS Development Event, Chengdu (2026) - ArkTS, Cangjie, DevEco, and AI-assisted development.
- 3rd Place, Douyin AI Hackathon, Chengdu (2026) - Team Paradox; Team Leader.
- Outstanding Student Leader, Neusoft Youth Club, Chengdu Neusoft University (2025).
- 10th National Biomedical Engineering Innovation Competition (2025) - Nuclear Radiation Detection, Xi'an Jiaotong University.
- Highest Mark & Award Nomination, Neusoft BloodBank WeChat Mini Program (2025) and VeriCart AI Project (2026).
- Ambassador & Innovation Contributor - ICT Olympiad Bangladesh (2024–Present) and Robotics Advisor - Robo Tech Valley (2025–Present)
- 8th RUSC National Science Festival, University of Rajshahi - Android Application Innovation and AI Business Model Project (2024–2025)
- Sichuan International Student Innovation & Entrepreneurship Competition (2024) - Battuta AI Travel Assistant.
- International Volunteer Member, Foreign Affairs Office of Dujiangyan City, Chengdu, China (2024)

LANGUAGES

Bengali: Native | English: Fluent | Hindi: Fluent | Urdu: Fluent | Chinese: Basic | Arabic: Elementary